

Investigating Gender Pay Gap in STEM Fields



ZELAN EROZ ESPANTO

SRAVYA MANDAVA

LAURA CHEN

NINJIN BILGEE

Project Introduction:

This study investigates the existence and extent of gender-based salary disparities within STEM fields, with a particular focus on how education level, major, and years of experience influence these gaps. Despite growing awareness and advocacy for equality, previous research suggests that a gender pay gap persists in many sectors, including STEM.

"Salary Prediction" Dataset from Kaggle

To investigate gender-based salary disparities in STEM fields, this study utilizes the "Salary Prediction for Beginner" dataset from Kaggle. This dataset provides detailed information on individuals' education levels, majors, years of professional experience, and annual salaries—key factors that commonly influence compensation.

- 6702 × 6 data
- Columns: Age, Gender, Education Level, Job Title, Years of Experience, and Salary

Procedure

1. Data pre-processing and cleaning
2. Exploratory Data Analysis (EDA)
3. Statistical Analysis (using p-value and Cohen's D)
4. Visualizations and Interpretations

Data Preprocessing

- Dropped rows with at least 1 null values.
- Standardized categorical variables
 - Ex. "high school" to "High School" and other variations
- Created bins for experience levels (by 5 years)
- Handled Salary Outliers using IQR (we found none)

Exploratory Data Analysis

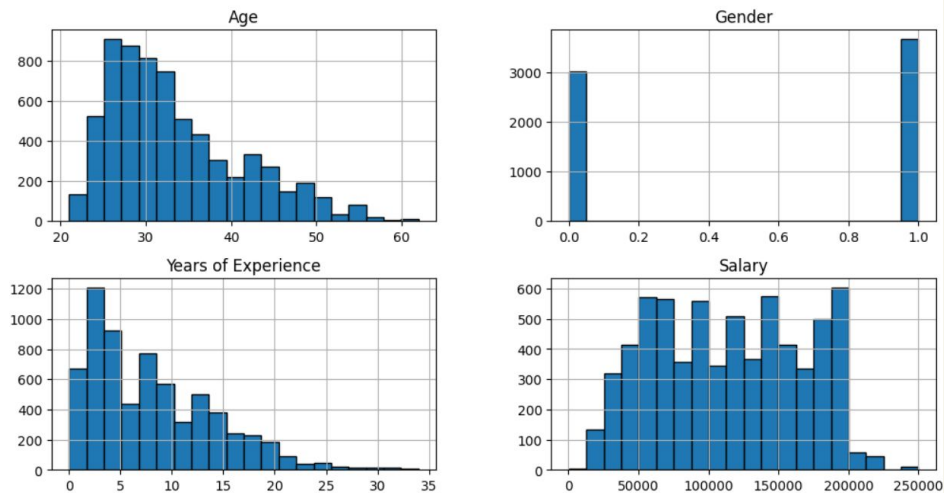
We presented this for our progress presentation. Through EDA we were able to look into:

- Distribution by Gender, Education Level, Age, and Job Titles
- Correlation Between Features

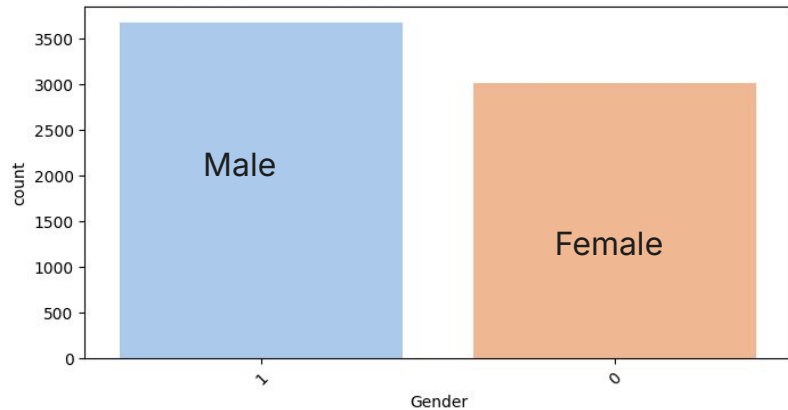
We realized it is difficult to look into the disparities given the 193 different job titles. From this we realized we could have the following options:

- Just look into the disparities among the top 5 jobs (based on distribution)
- Categorize the given 193 unique job titles into industry categories (ex. finance, management, tech, etc) and conduct the comparison within these industry categories

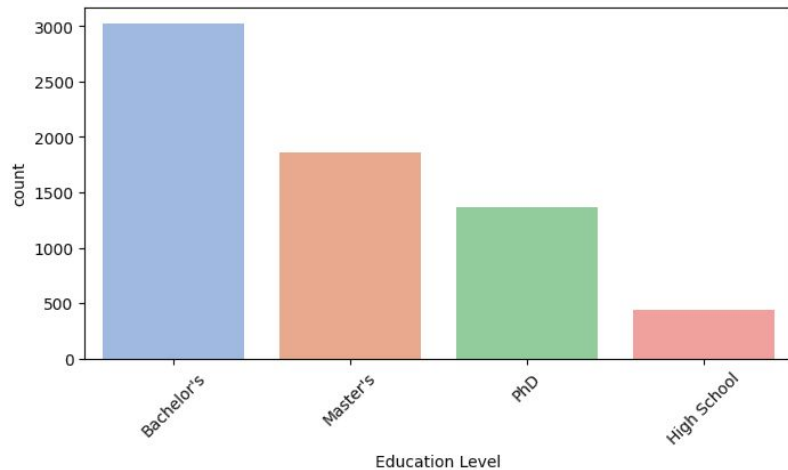
Distribution of Numerical Features



Distribution of Gender



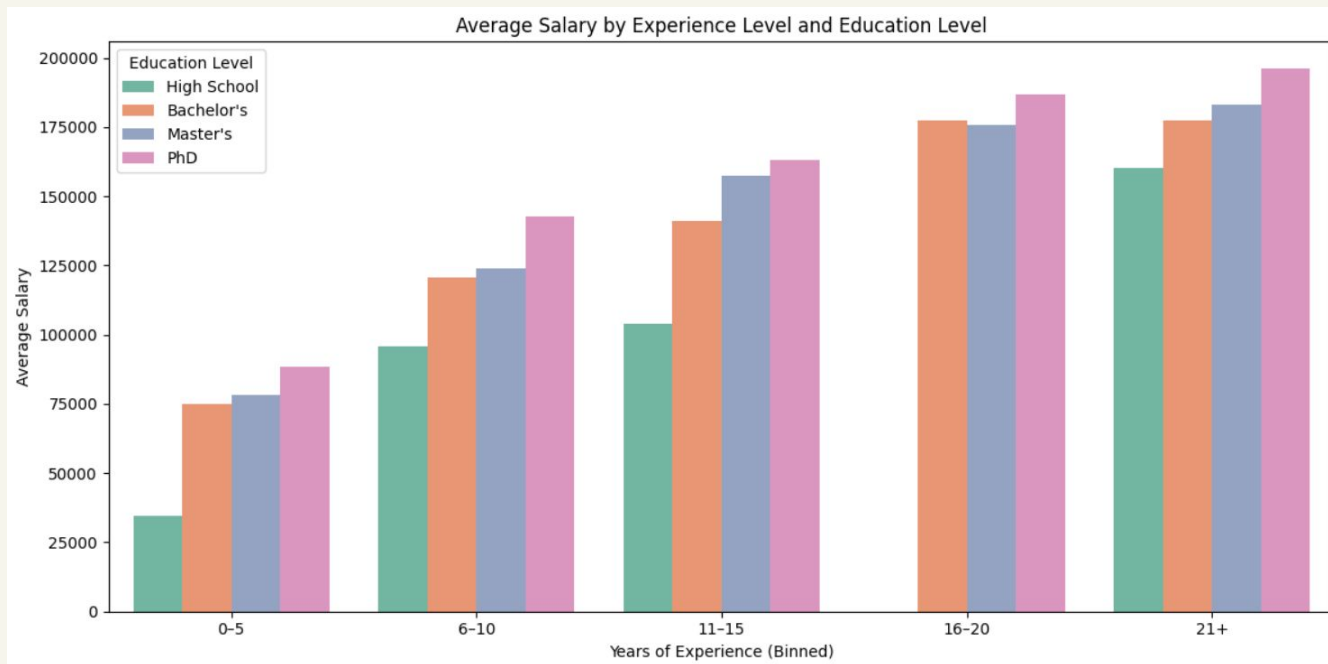
Distribution of Education Level

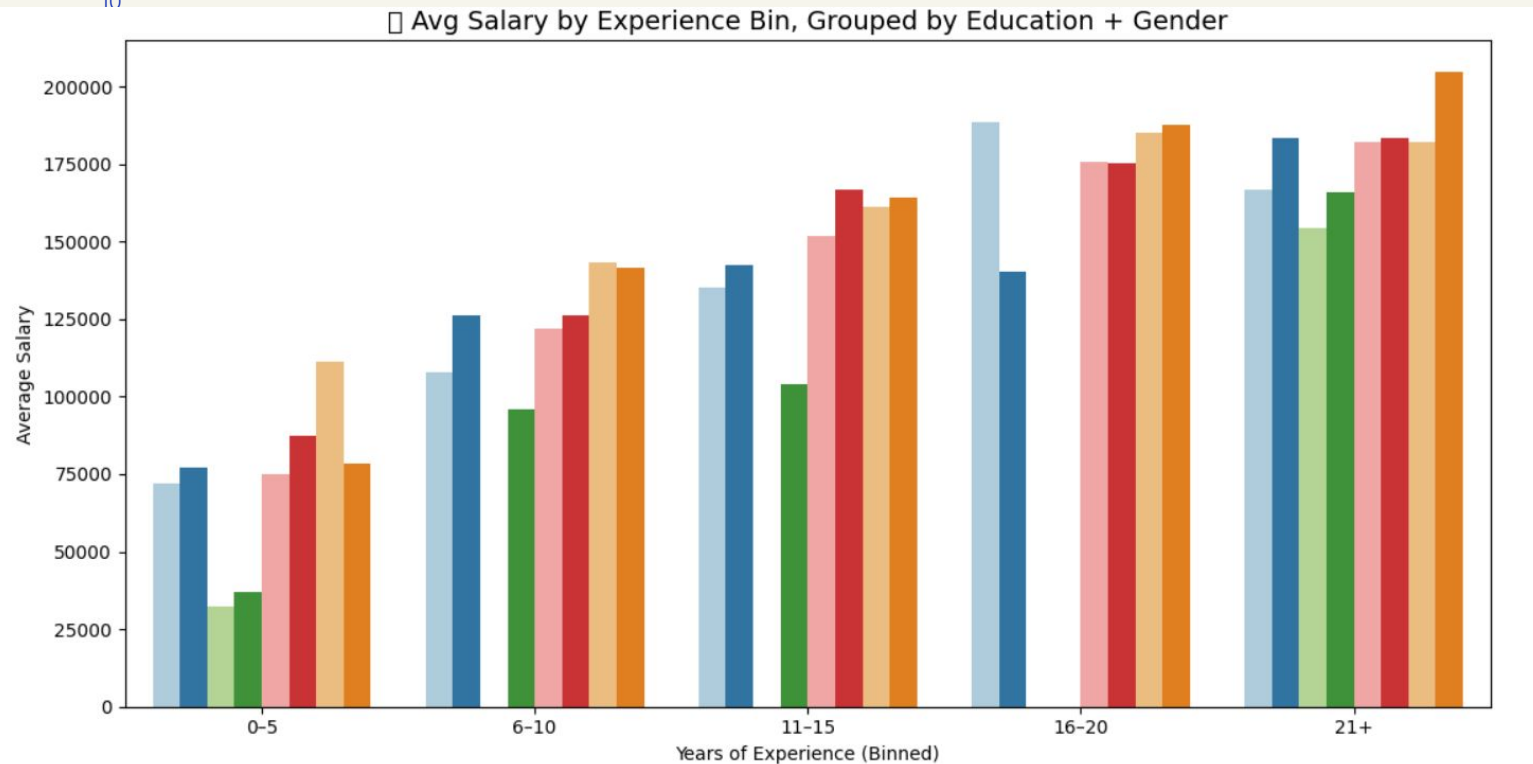


PART I.

By Education Level and Years of Experience

COMPARING THE DIFFERENT SALARY RANGES FOR DIFFERENT EDUCATION LEVELS



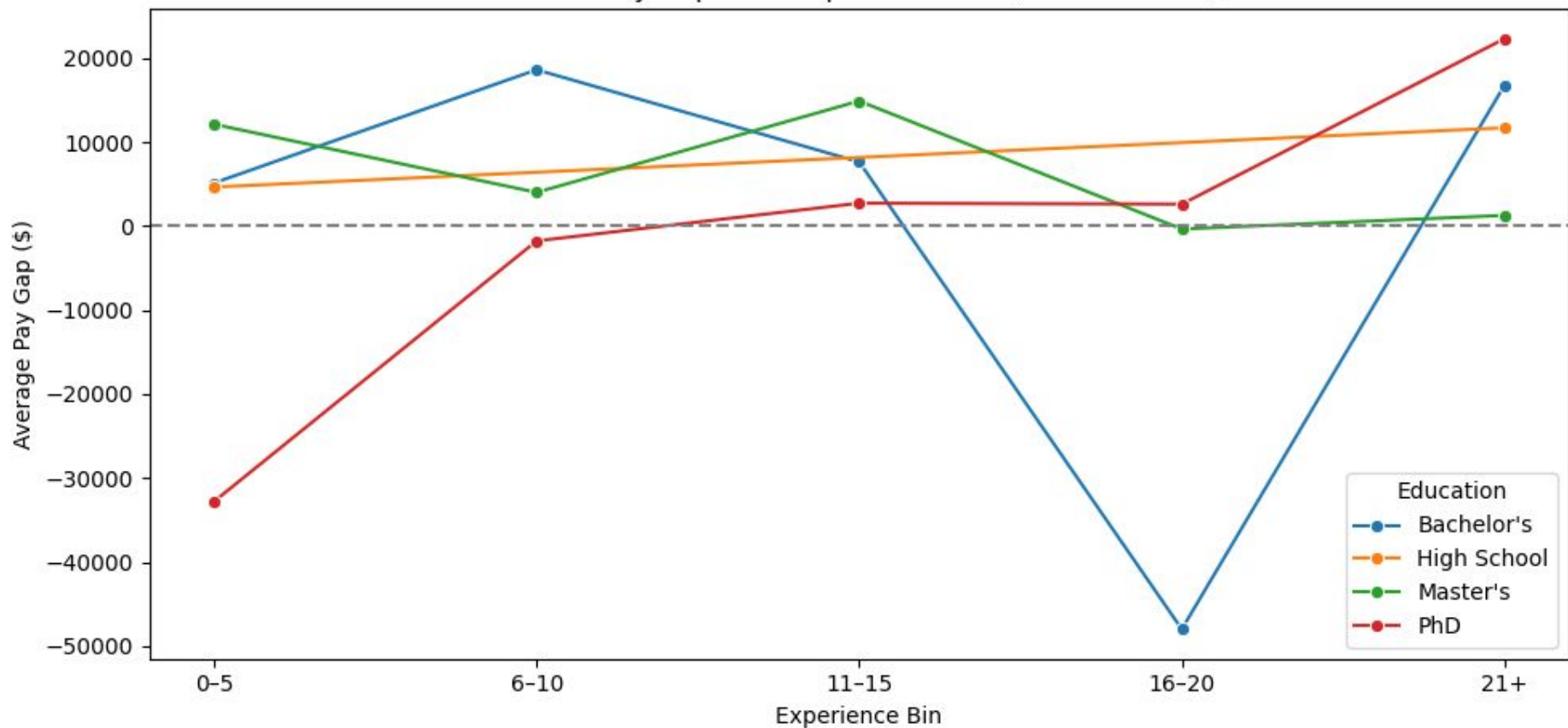


Education Level + Gender

- Bachelor's - Female
- Bachelor's - Male
- High School - Female
- High School - Male
- Master's - Female
- Master's - Male
- PhD - Female
- PhD - Male

Gender Bin	Experience Bin	Education	Female Salary	Male Salary	Pay Gap (M - F)	Gender % Gap
0	0-5	Bachelor's	72101.49	77234.93	5133.44	7.12
1	0-5	High School	32210.9	36862.17	4651.26	14.44
2	0-5	Master's	75194.29	87326.22	12131.93	16.13
3	0-5	PhD	111165.05	78376.41	-32797.41	-29.5
4	6-10	Bachelor's	107805.68	126422.12	18616.44	17.27
5	6-10	High School		95845.0		
6	6-10	Master's	121992.17	126007.54	4015.38	3.29
7	6-10	PhD	134967.95	141708.76	-1759.19	-1.23
8	11-15	Bachelor's	134974.9	142618.34	7643.44	5.66
9	11-15	High School		103947.0		
10	11-15	Master's	151779.68	166675.86	14896.18	9.81
11	11-15	PhD	161385.58	164118.04	2732.46	1.69
12	16-20	Bachelor's	188403.2	140415.86	-47987.35	-25.47
13	16-20	Master's	175814.5	175469.73	-344.77	-0.2
14	16-20	PhD	184962.57	187520.75	2620.18	1.42
15	16-20	Bachelor's	166621.28	183371.62	16750.35	10.05
16	21+	High School	154207.0	165919.0	11712.0	7.59
17	21+	Master's	182166.82	183443.06	1276.25	0.7
18	21+	PhD	182297.45	204610.91	22313.46	12.24

Gender Pay Gap Over Experience Bins (Male - Female)



Index Gender Experience Bin Education Female Salary Male Salary Pay Gap (M-F) % Gap

0	0	0-5	Bachelor's	72,101.49	77,234.93	5,133.44	7.12
1	0	0-5	High School	32,210.90	36,862.17	4,651.26	14.44
2	0	0-5	Master's	75,194.29	87,326.22	12,131.93	16.13
3	0	0-5	PhD	111,165.05	78,367.64	-32,797.41	-29.50
4	0	6-10	Bachelor's	107,805.68	126,422.12	18,616.44	17.27
5	0	6-10	High School	NaN	95,845.00	NaN	NaN
6	0	6-10	Master's	121,992.17	126,007.54	4,015.38	3.29
7	0	6-10	PhD	143,467.95	141,708.76	-1,759.19	-1.23
8	0	11-15	Bachelor's	134,974.90	142,618.34	7,643.44	5.66
9	0	11-15	High School	NaN	103,947.00	NaN	NaN
10	0	11-15	Master's	151,779.68	166,675.86	14,896.18	9.81
11	0	11-15	PhD	161,385.58	164,118.04	2,732.46	1.69
12	0	16-20	Bachelor's	188,403.20	140,415.86	-47,987.35	-25.47
13	0	16-20	Master's	175,814.50	175,469.73	-344.77	-0.20
14	0	16-20	PhD	184,962.57	187,582.75	2,620.18	1.42
15	0	21+	Bachelor's	166,621.28	183,371.62	16,750.35	10.05
16	0	21+	High School	154,207.00	165,919.00	11,712.00	7.59
17	0	21+	Master's	182,166.82	183,443.06	1,276.25	0.70
18	0	21+	PhD	182,297.45	204,610.91	22,313.46	12.24

High School:

- Gaps are generally positive (males earn more).
- Limited data (many NaNs).
- Highest gap at 21+ years: **\$11,712 (7.59%)**.

Bachelor's

- Consistently positive gaps.
- Largest gap at 21+ years: **\$16,750, (10.05%)**
- Exception: 16-20 years shows **reverse gap** of **\$-47,987, -25.47%**

Master's

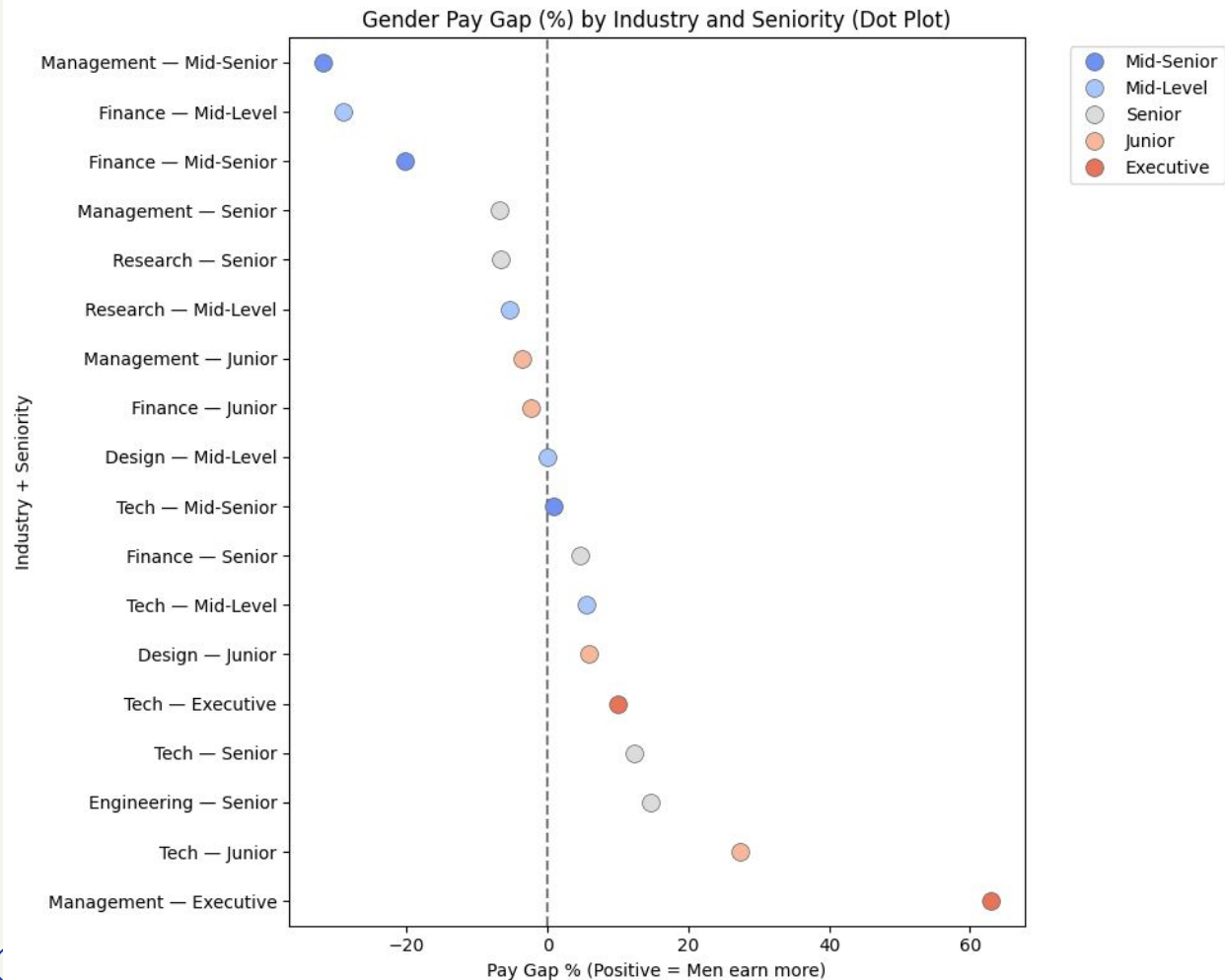
- Generally male-dominant gaps but **lower %** than others.
- Gap is smallest (almost 0) at 16-20 years.
- Slight negative gap at 6-10 years.

PhD

- Mixed results.
- At 0-5 years, females earn **more (-29.50% gap)**.
- At 21+ years, males earn **\$22,313 more, 12.24% gap**

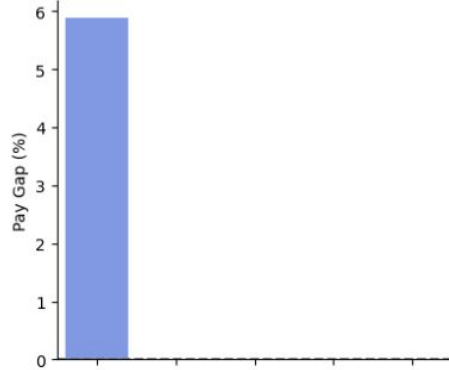
PART II.

By Industry and Seniority Level

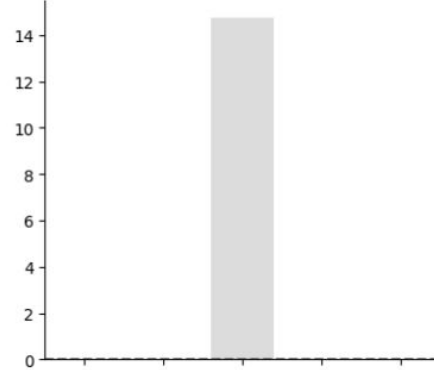


Gender Pay Gap (%) by Seniority Level (per Industry)

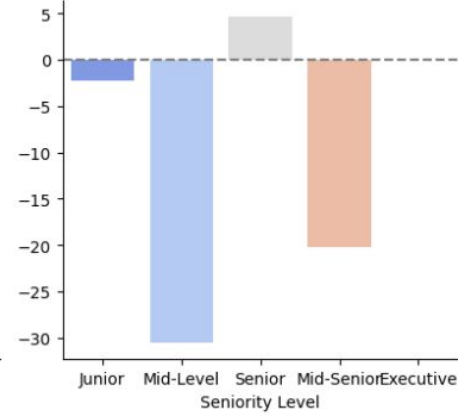
Design



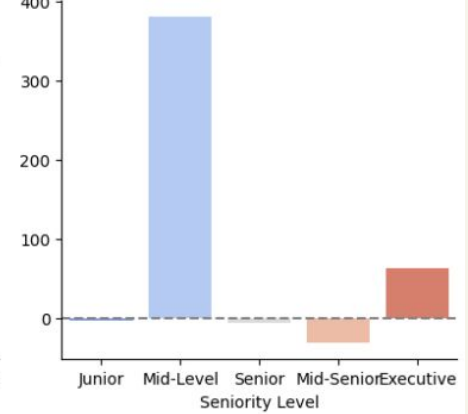
Engineering



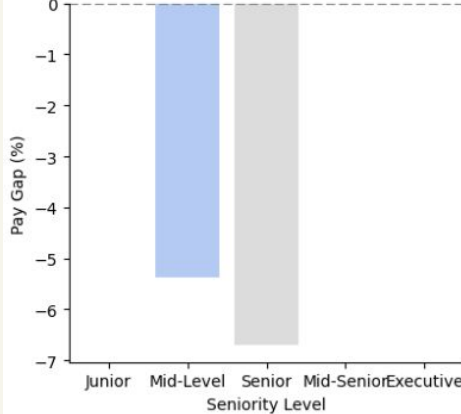
Finance



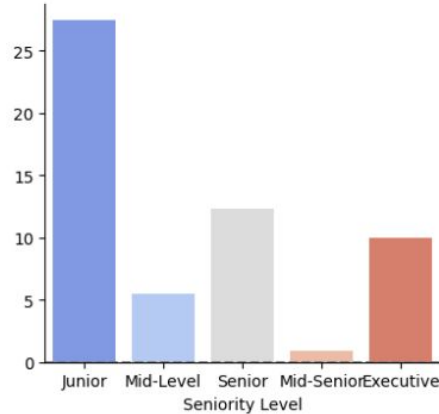
Management



Research



Tech



Design

- Only the Junior level shows a gap (~6%) against women.
- All other levels show no gap.

Engineering

- Only Senior level is shown, with a ~14% gap — females earn less.

Finance

- Mid-Level has the most extreme negative gap (~-30%) → women earn more than men.
- Senior level is balanced.
- Executive level again favors men (~-20%).

Management

- Mid-Level shows an unusually high gap (~380%) — possibly data error or extreme outlier?
- Senior and Mid-Senior show small gaps, even negative.
- Executive levels favor women (~60%).

Research

- Mid-Level and Senior show reverse gaps (~-5% and -6%).
- Women earn slightly more in these roles.

Tech

- Gaps exist at all levels, though smaller than other industries:
 - Junior: ~27%
 - Mid-Level to Executive: Between 1% and 10%

	Industry	Seniority	Male Mean Salary	Female Mean Salary	Mean Difference	p-value	Cohen's d
3	Tech	Junior	50792.207792	39861.386139	10930.821654	3.805801e-33	2.257766
4	Management	Mid-Senior	116238.372093	170703.703704	54465.331611	4.908979e-27	-1.380693
2	Tech	Senior	156469.307317	139294.109091	17175.198226	6.0440510e-07	0.631786
5	Research	Senior	132857.142857	142400.000000	-9542.857143	1.989944e-06	-1.549972
7	Engineering	Senior	173188.553488	150938.980583	22249.572906	1.235134e-05	0.586285
1	Tech	Executive	161151.079137	146475.903614	14675.175522	1.427316e-04	0.384100
6	Finance	Mid-Senior	119677.419355	150097.087379	30419.668024	5.286115e-04	-0.616114
0	Tech	Mid-Level	120529.771549	114228.157279	6301.614269	6.093858e-04	0.142788

PART III.

Looking into the Potential of a Salary Prediction Model



Regression Model Performance Comparison:

	Model	RMSE	R^2 Score
4	XGBoost	18330.170987	0.869234
3	Random Forest	18622.758632	0.865026
1	Ridge Regression	28631.035088	0.680966
2	Lasso Regression	28639.271051	0.680782
0	Linear Regression	28640.429667	0.680756
5	Support Vector Machine	50198.173018	0.019293

```
✓ [104] # Load the model
0s model = joblib.load('xgb_salary_pipeline.pkl')

# Example user input
user_input = pd.DataFrame([{'Gender_Label': 'Female',
                             'Education Level': "Bachelor's",
                             'Seniority Level': 'Senior',
                             'Job Industry Category': 'Tech',
                             'Years of Experience': 5
                           }])

# Predict
predicted_salary = model.predict(user_input)[0]
print(f"💰 Predicted Salary: ${predicted_salary:,.2f}")
```

🔄 💰 Predicted Salary: \$104,347.49

PART IV.

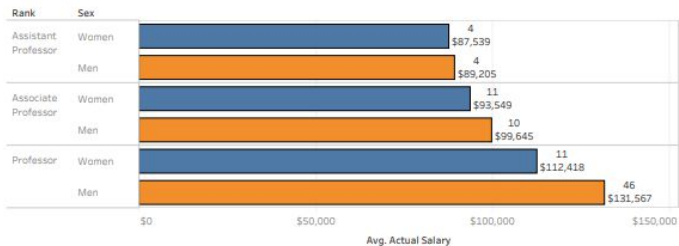
Looking at CWRU Faculty

Faculty Salary Analysis

Methodology and Analysis	Preface to the Results	College of Arts and Sciences: Arts and Humanities	College of Arts and Sciences: Math and Natural Sciences	College of Arts and Sciences: Social Sciences	Case School of Engineering	Mandel School of Applied Sciences
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Average Salary by Sex and Rank: Math and Natural Sciences

School/Division
Math and Natural Scienc..



Above or Below the Line

Sex	Above or Below	Percentage
Women	Above the line	42%
	Below the line	58%
Men	Above the line	50%
	Below the line	48%
	On the line	2%

Average Residuals



- 58% of women are below the average salary
- 48% of men are below the average salary
- **Men:** The average residual is **+\$2,057**
- **Women:** The average residual is **-\$4,746**

Conclusions

- Bachelor's and High School male salaries peak earlier but plateau or even drop in later experience bins — a unique trend not mirrored by females.
- Overall, gender pay gaps are most visible in early-career stages, especially at lower education levels, and tend to decrease or reverse as experience increases.
- Gender pay gaps exist across nearly every industry and seniority level, though Research and Design show more balanced distributions.
- Small sample sizes in some groups (e.g., Executive–Design) highlighted the importance of minimum group size thresholds in significance testing.
- Among 6 models tested, XGBoost achieved the best performance, with the lowest RMSE and highest R^2 score.
- Linear, Ridge, and Lasso regressions performed similarly, suggesting the relationships aren't strictly linear.

Next Steps

- Expand the dataset to include more industries, job roles, race/ethnicity (if ethical & available), company size, and geographic region.
- Control for outliers more rigorously — set minimum data thresholds per group (e.g., $N \geq 5$) and use IQR/z-score filtering.
- Explore longitudinal data (pay progression over time) and conduct promotion gap analysis.
- Partner with HR or policy teams to translate findings into equitable pay structures and compensation audits.
- Include Equity Metrics in Company Performance Reviews.

References

- Faculty Salary Analysis for CWRU from 2020-2021
<https://case.edu/ir/sites/default/files/2021-03/Faculty%20Salary%20Analysis%202020.pdf>
- DATA SET REFERENCE:
- <https://www.kaggle.com/datasets/rkiattisak/salaly-prediction-for-beginner>

Questions?